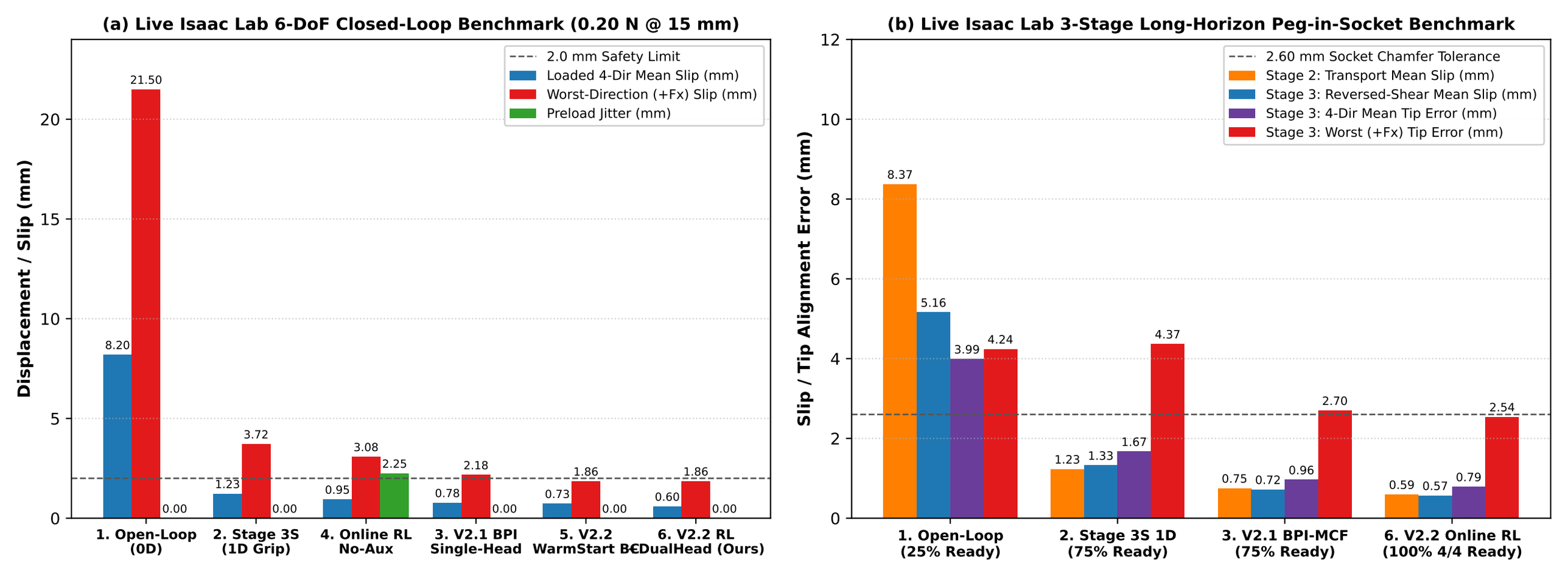
(a) Live Isaac Lab 6-DoF Closed-Loop Benchmark (0.20 N @ 15 mm)
Displacement / Slip (mm)
2.0 mm Safety Limit
Loaded 4-Dir Mean Slip (mm)
Worst-Direction (+Fx) Slip (mm)
Preload Jitter (mm)
8.20
21.50
0.00
1.23
3.72
0.00
0.95
3.08
2.25
0.78
2.18
0.00
0.73
1.86
0.00
0.60
1.86
0.00
1. Open-Loop (0D)
2. Stage 3S (1D Grip)
4. Online RL No-Aux
3. V2.1 BPI Single-Head
5. V2.2 WarmStart B€
6. V2.2 RL DualHead (Ours)
(b) Live Isaac Lab 3-Stage Long-Horizon Peg-in-Socket Benchmark
Slip / Tip Alignment Error (mm)
2.60 mm Socket Chamfer Tolerance
Stage 2: Transport Mean Slip (mm)
Stage 3: Reversed-Shear Mean Slip (mm)
Stage 3: 4-Dir Mean Tip Error (mm)
Stage 3: Worst (+Fx) Tip Error (mm)
8.37
5.16
3.99
4.24
1.23
1.33
1.67
4.37
0.75
0.72
0.96
2.70
0.59
0.57
0.79
2.54
1. Open-Loop (25% Ready)
2. Stage 3S 1D (75% Ready)
3. V2.1 BPI-MCF (75% Ready)
6. V2.2 Online RL (100% 4/4 Ready)